

CS 486/686

Pretraining Language Models From First Principles

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Lecture 20

How raw text becomes a training signal

Search ›

Uncertainty ›

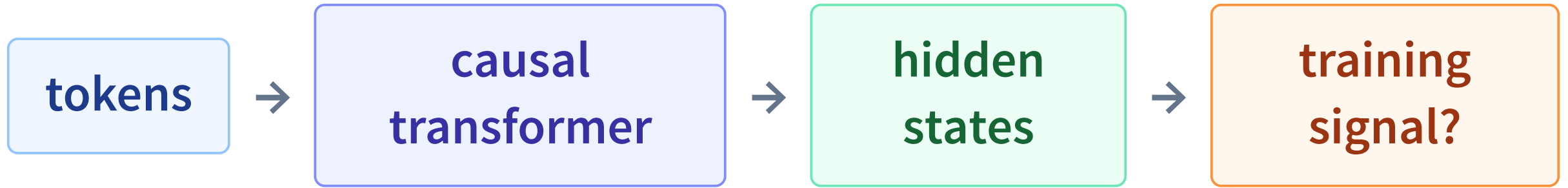
Decisions ›

Learning

By the end, you can trace one training example

- Turn text into **tokens, IDs, and vectors**.
- Build **next-token targets** from one sequence.
- Trace $\mathbf{h}_t \rightarrow \text{logits} \rightarrow \text{probability} \rightarrow \text{loss}$.
- Explain how one batch **updates every parameter**.
- Explain why targets are automatic but **data curation is not**.
- Contrast parallel **training** with sequential **generation**.

L19 built the machine. What teaches it?



Give it one task, billions of times: **predict what comes next.**

A language model predicts a distribution

The robot picked up the cup because it was ____

empty

very plausible

blue

also plausible

broken

possible

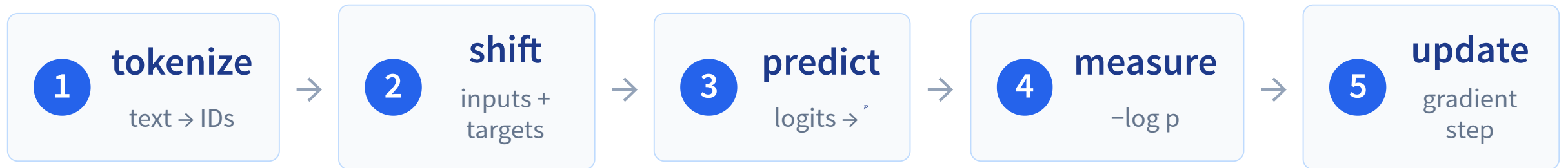
quantum

unlikely

training text continues with **empty**

The label is the *observed* next token—not the only meaningful continuation.

Our roadmap: text becomes a gradient



Tokens are not words

hello $\neq$ ·hello

a leading space matters

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rare strings split

unbelievable $\rightarrow$ un | belie | vable

pieces need not be morphemes

  $\rightarrow$ tokens

bytes let any Unicode text be represented

Token count controls sequence length—and therefore memory and compute.

Learn reusable pieces by merging frequent pairs

start with characters

l | o | w | e | r

l | o | w | e | s | t

merge l + o

lo | w | e | r

lo | w | e | s | t

after many merges

low | er

low | est

Save the learned pieces: the tokenizer can now emit **low**, **er**, **est**, ... as individual tokens.

Freeze this piece list, then use the same
tokenizer for every language-model example.

Byte-pair encoding intuition; Sennrich, Haddow & Birch, “Neural Machine Translation of Rare Words with Subword Units,” 2016.

A real tokenizer, piece by piece LIVE

Qwen3 tokenizer · 151,669 entries · · means “this token begins with a space.”

unbelievable

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hello

·hello

code

unbelievable

tokenize custom text

un
359

belie
31798

vable
23760

12 characters **12** UTF-8 bytes **3** tokens

Separate chips are separate tokens. The · symbol is not a separator; it represents one leading blank.

IDs retrieve learned vectors

token	ID	lookup
The	785	$\mathbf{E}_{785}$
· robot	12305	$\mathbf{E}_{12305}$
· empty	4287	$\mathbf{E}_{4287}$

IDs are addresses.

Nearby numbers need not mean nearby concepts.

Embedding rows are learned.

Pretraining gives the vectors meaning.

embedding vectors

with

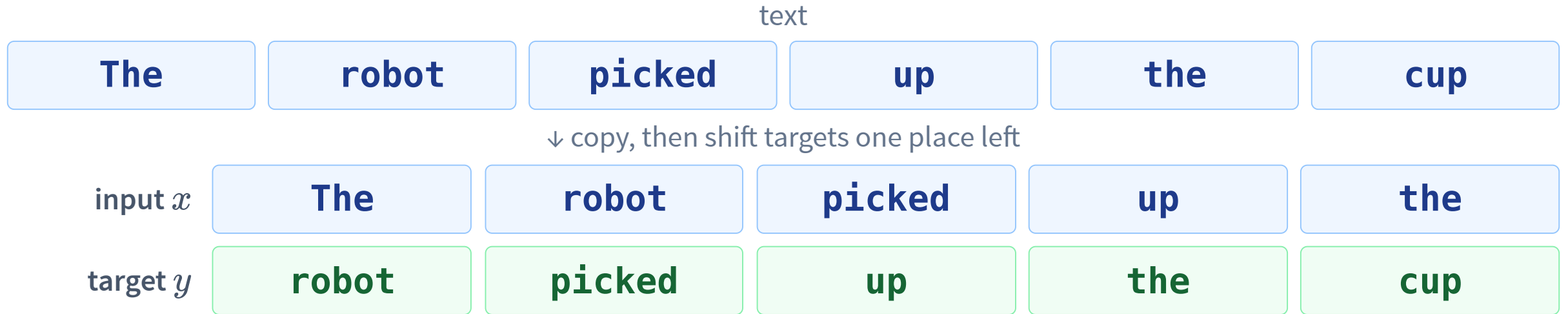
positional information

→

causal transformer

Actual IDs from the Qwen3 tokenizer; vocabulary size 151,669.

Shift once: one sequence gives many labels



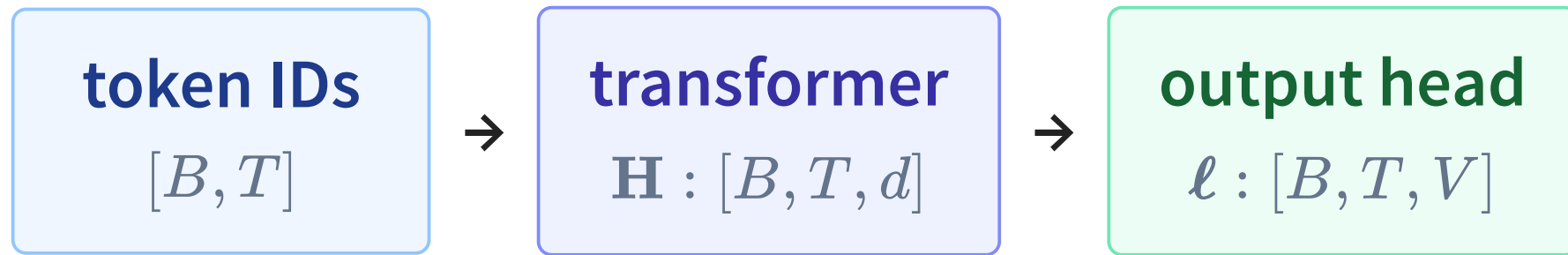
Five observed tokens become five classification labels—no manual annotation.

Teacher forcing predicts all positions together

position	1	2	3	4	5
given token	The	robot	picked	up	the
predict	robot	picked	up	the	cup
may read	1	1–2	1–3	1–4	1–5

One forward pass: all true prefixes are present; the causal mask blocks every future token.

Each hidden state votes over the vocabulary



$$\ell_t = \mathbf{W}_{\text{out}} \mathbf{h}_t + \mathbf{b}$$

At every position: V scores—one for every possible next token.

Logits are unnormalized preferences

TOY VOCABULARY

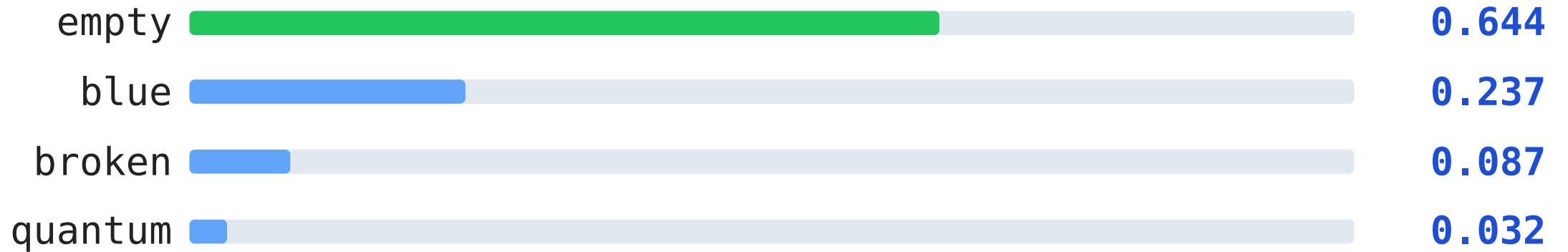
context: The robot picked up the cup because it was

candidate	logit	meaning
empty	2.0	observed target
blue	1.0	second choice
broken	0.0	less preferred
quantum	-1.0	least preferred

Logits can be negative and need not sum to anything.

Softmax turns those scores into probabilities

$$p_i = \frac{e^{l_i}}{\sum_j e^{l_j}}$$



$$0.644 + 0.237 + 0.087 + 0.032 = 1.000$$

The observed token determines the loss

observed next token

empty

model assigned

$$p(\text{empty}) = 0.644$$

cross-entropy

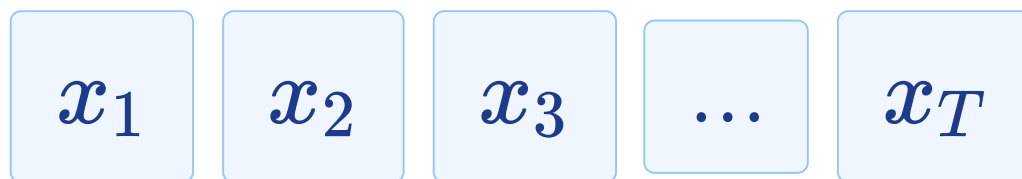
$$-\ln 0.644 = 0.44$$

$$p(\text{target}) = 0.90 \Rightarrow L = 0.11$$

$$p(\text{target}) = 0.01 \Rightarrow L = 4.61$$

Confident and wrong is expensive.

A text probability is built left to right



$$P(x_1, \dots, x_T) = P(x_1) \prod_{t=1}^{T-1} P(x_{t+1} \mid x_{\leq t})$$

$$P(\text{The}) \times P(\text{robot} \mid \text{The}) \times P(\text{picked} \mid \text{The robot}) \times \dots$$

The same conditional probabilities power both training and generation.

Training averages the token losses

$$\mathcal{L}(\theta) = -\frac{1}{N} \sum_{i=1}^N \log p_{\theta}(y_i \mid \text{context}_i)$$

N

non-padding targets in the batch

lower is better

on held-out validation text

perplexity

$\exp(\mathcal{L}_{\text{val}})$

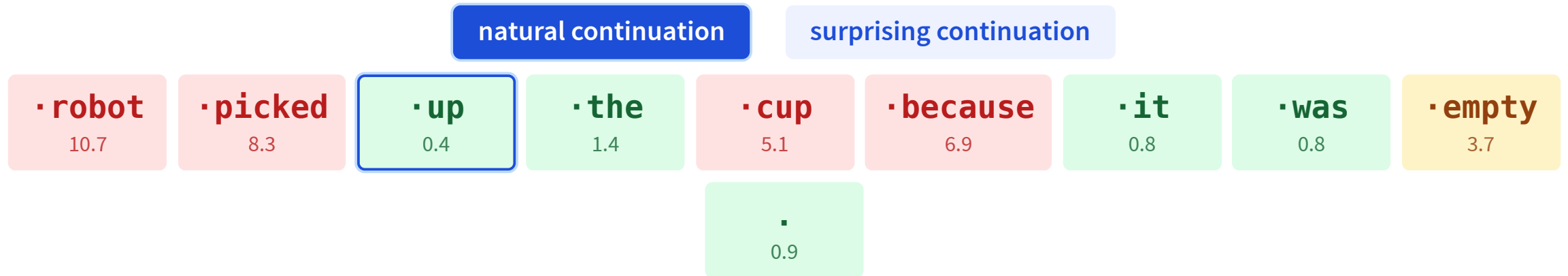
If $\mathcal{L}_{\text{val}} = 2$, perplexity is $e^2 \approx 7.4$: uncertainty like roughly seven equally likely choices.

Inspect a real model's loss

LIVE DATA

Qwen3-0.6B-Base · click a target token to inspect its prefix, probability, and loss.

Color encodes target loss: green = lower, amber = medium, red = higher. Numbers are nats.



prefix

The robot picked

target · up

p = 0.668

-ln p = 0.40

predictable here → small gradient signal

average NLL 3.90

perplexity 49.2

10 training signals

Click any target token. Its prefix is exactly what that position could use under the causal mask.

The pretraining loop, line by line

```
# packed token sequences: one extra token supplies the final target
```

```
ids = next(token_batches)          # [B, T+1]
```

```
# every position predicts the token immediately to its right
```

```
x, y = ids[:, :-1], ids[:, 1:]     # each [B, T]
```

```
# one vocabulary-sized score vector at every position
```

```
logits = model(x).logits           # [B, T, V]
```

```
# average next-token cross-entropy over the whole batch
```

```
loss = cross_entropy(logits.reshape(-1, V), y.reshape(-1))
```

```
# discard gradients left from the previous update
```

```
optimizer.zero_grad()
```

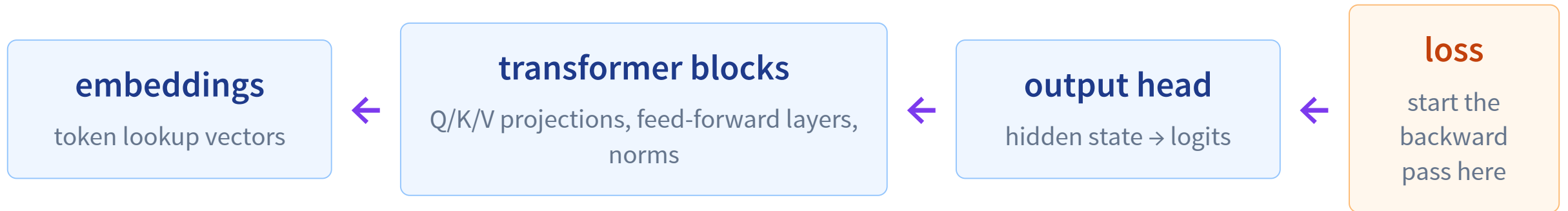
```
# autograd sends credit and blame through the entire model
```

```
loss.backward()
```

```
# update all trainable parameters once
```

```
optimizer.step()
```

The gradient reaches the whole model



Important: attention *weights* are recomputed for each input; the projection matrices that produce them are learned parameters.

Self-supervised does not mean “free”

Targets are automatic

The robot picked → up

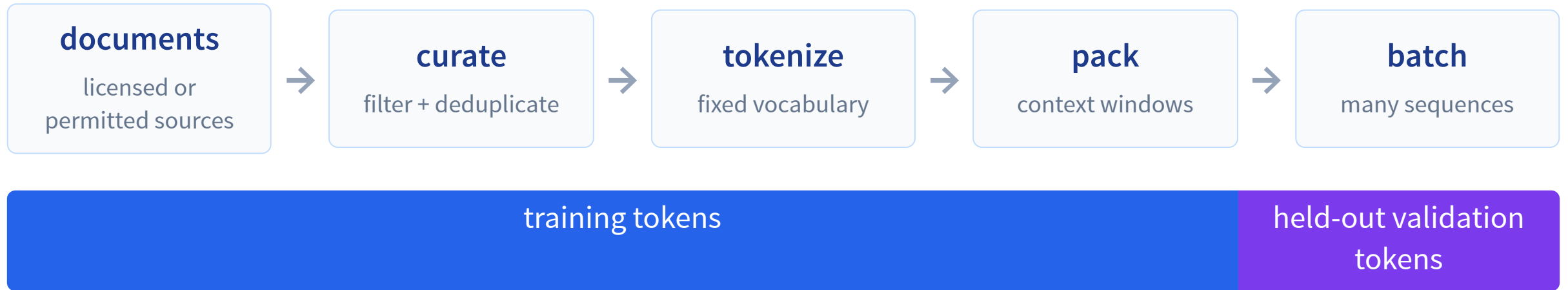
No person labels every next token.

Data is not automatic

- permission and licensing
- filtering and deduplication
- privacy, safety, and quality

The objective is cheap to label; the corpus is costly to build well.

Before training: build the token stream



Never evaluate next-token loss on text used for parameter updates.

Scale is a budget: balance model and data

too many parameters

model size

data

too few tokens to train them well

balanced for fixed compute

model size

training tokens

parameters and tokens grow together

rough training compute $\propto$ parameters $\times$ training tokens

Hoffmann et al., “Training Compute-Optimal Large Language Models,” 2022.

What can next-token prediction teach?

syntax + style

The keys `are` on the table.

associations

Waterloo is in `Ontario`.

procedures

```
for i in range(3): print(i)
```

To predict well, hidden states must capture many recurring patterns in text and code.

But the objective guarantees only prediction

truth

frequent text can still be false

reasoning

pattern completion can fail off-distribution

instructions

a base model is trained to continue, not obey

recency

weights do not know events after training

privacy

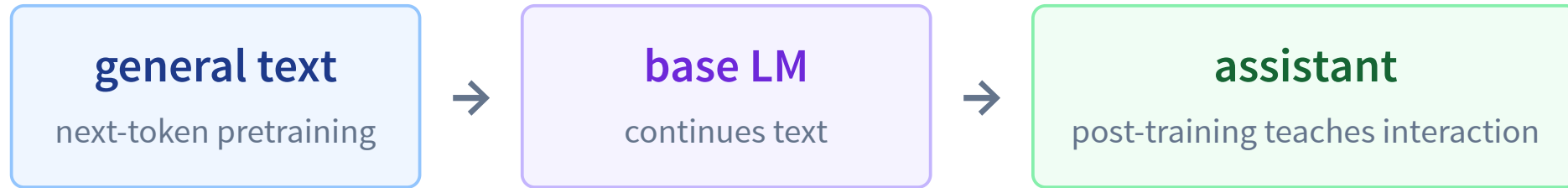
training text may be memorized

new context

private/current facts must be supplied safely

Low loss is useful—not a certificate of truth or helpfulness.

Pretraining creates a base language model



Explain gradient descent:

a base model may continue the document—not necessarily follow the request.

Later: instruction tuning, preference optimization, prompting, RAG, and tools.

Same model, different execution

training

true sequence → all positions → loss → update weights

- true prefixes supplied
- many token predictions in parallel

generation

prompt → one distribution → choose + append ↺

- weights fixed
- new tokens arrive sequentially

Choose one token, append, repeat

REAL MODEL DATA

Qwen3-0.6B-Base · choose a rule, then click repeatedly to grow the context one token at a time.

Shakespeare

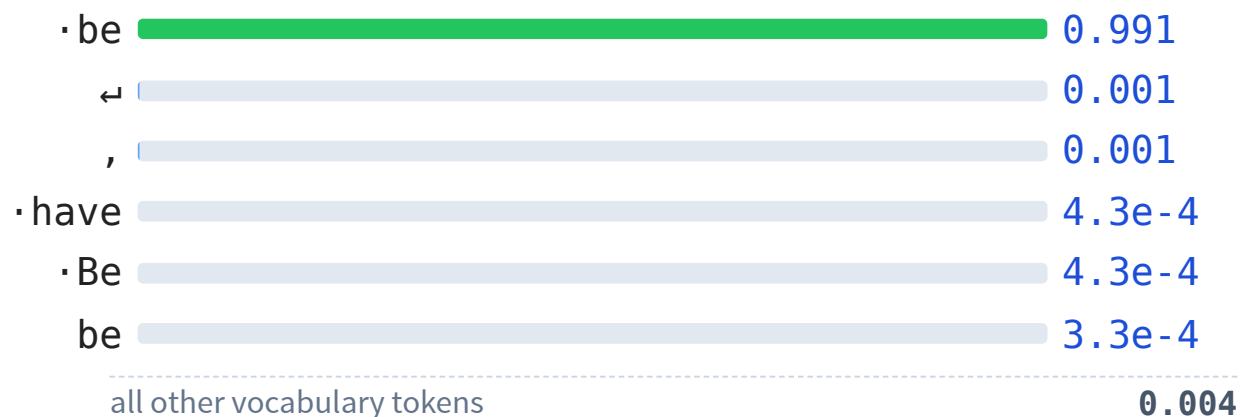
robot

story

step 1: To be, or not to → ?

highest each step

sample each step



append next token

reset

Click “append next token” repeatedly. Each click uses the distribution conditioned on the enlarged context.

To be, or not to

In candidate labels, · means a leading space; the appended context renders the real blank. Every click then updates to the next conditional distribution.

Next: watch Qwen run from the inside

inspect

tokenizer, weights, attention

generate

append loop + KV cache

control

greedy, sampling, temperature, top-k/top-p

condition

chat templates and prompts

L21: Dissecting Qwen3-0.6B.

Exit check: can you trace the signal?

The	robot	picked
robot	picked	up

At the final position, the model assigns $p(\text{up}) = 0.80$.

target up	token loss $-\ln 0.80 \approx 0.22$
training average with other positions, then backpropagate	generation choose a token, append it, repeat

Text $\rightarrow$ targets $\rightarrow$ probabilities $\rightarrow$ loss $\rightarrow$ gradients.